

ANSWERS

1.

Question	Answer	Marks
3(a)	particles move apart or distance between particles increases	B1
	particles able to move further against the forces (of attraction) or forces (of attraction) less effective against faster moving particles or particles collide harder (with neighbours)	B1
3(b)(i)	conduction	B1
3(b)(ii)	<u>density</u> of heated water decreases	B1
	less dense / heated water rises	B1
	cold water sinks / convection current established	B1
3(b)(iii)	100 °C or 373 K	B1
3(b)(iv)	$(\Delta T =) 100 - 17$ or 83 (°C)	C1
	$(\Delta E =) mc\Delta T$ or $2.5 \times 4200 \times (100 - 17)$ or $2.5 \times 4200 \times 83$	C1
	8.7×10^5 (J)	A1

2.

Question	Answer	Marks
3(a)	chemical potential energy and no other energy	B1
3(b)(i)	radiation or infrared mentioned	B1
	radiation and infrared (travel to the man)	B1
	energy / infrared / radiation <u>absorbed</u> by the man / pullover or travels in straight line or travels at the speed of light or increases (man's) internal energy	B1
3(b)(ii)	(shiny surface) <u>reflects</u> radiation (to man)	B1
	less (thermal) energy escapes or more (thermal) energy reaches / absorbed by man	B1
3(b)(iii)	EITHER black (surface) is good absorber / poor reflector of infrared radiation	B1
	more (thermal) energy absorbed or more thermal energy transferred to man	B1
	OR trapped air and reduced convection / air poor conductor / air is a good insulator	(B1)
	less (thermal) energy lost	(B1)

3.

Question	Answer	Marks
2(a)	dull black line 100	B1
	dull white 60 and shiny white 40	B1
2(b)	hot air rises	B1
	density of hot air less (than cold air) or air expands on heating	B1
	hot air rises <u>and</u> <u>cold</u> air falls or a complete current described	B1

4.

Question	Answer	Marks
3(a)(i)	water expands (as it is heated) or molecules move apart	B1
	density (of heated water) decreases or (heated water) rises	B1
	it rises and the cold water falls or convection current established	B1
3(a)(ii)	water (below X) heated by conduction or no convection occurs	C1
	water is a poor conductor	A1
3(b)	air / plastic is a poor / slow (thermal) conductor	B1
	trapping the air prevents convection (in air) or no <u>free</u> electrons (in plastic)	B1

5.

5(b)(i)	(microwaves) must pass through or not be absorbed (by container) or not reflected	B1
5(b)(ii)	convection and/or conduction named	B1
	(convection involves) hot liquid rises (cold liquid falls)	B1
	(conduction involves) energy passes from molecule to molecule	B1

6.

Question	Answer	Marks
4(a)	glass molecules vibrate	B1
	molecules pass on energy / vibration from one to another	B1
	contains <u>many / more</u> free electrons / electrons that can move	B1
	electrons collide with atoms / molecules (and gain / lose energy)	B1
4(b)(i)	glass warms up (before water) or glass expands first	B1
	water expands more than glass	B1
4(b)(ii)	convection or hot water rises	B1
	convection <u>current</u> formed or hot water rises and cold water falls or density of water falls (as it heats)	B1

7.

Question	Answer	Marks
9(a)	it is less dense (than the cooler water)	B1
	it floats on the cooler water / cooler water cannot move up / cooler water remains on the bottom	B1
9(b)(i)	copper / metal is a good (thermal) conductor	B1
9(b)(ii)	vibrating atoms / ions / particles / molecules or electrons gain energy	B1
	atoms / ions / particles / molecules hit the electrons or electrons travel (a long distance through the copper)	B1
	electrons hit / transfer energy to (distant) atoms / ions / particles / molecules	B1

9(b)(iii)	it contracts or its density increases	B1
	it sinks	B1
	less dense / warmer water rises or sets up a convection current	B1
9(c)	14(°C) or 0.25 (kg) or 250 / 1000 (kg) seen	C1
	$(Q =) mc\Delta t$ or $0.25 \times 4200 \times (21 - 7)$ or $0.25 \times 4200 \times 14$	C1
	14 700 J or 15 000 J	A1
9(d)(i)	any two from: molecules / they move in clusters	B1
	slide over each other molecules / they move throughout the liquid	B1
9(d)(ii)	average speed / <u>kinetic</u> energy decreases	B1

8.

Question	Answer	Marks
10(a)	<i>metal tip</i> 1000 °C	B1
	<i>solder</i> 200 °C	B1
10(b)(i)	$(E =)VIt$ algebraic or numerical	C1
	790 J	A1
10(b)(ii)	$(Q =) mcT$ algebraic or numerical	C1
	$2.3 \times 300 \times 0.39$	C1
	270 J	A1
10(c)(i)	electrons <u>move</u> around or electrons diffuse through (metal)	M1
	pass on energy to other parts / hit atoms or other electrons / (transfer energy) in all directions	A1
10(c)(ii)	hot air / heated air rises	B1
	(hot air) less dense (than cold air)	B1
10(d)(i)	two different metals	B1
	complete circuit with meter in series and one junction labelled H on hot tip	B1
10(d)(ii)	ANY TWO from - can measure rapid temperature changes - can measure high temperatures - is small in size - has a small heat capacity - can give electrical signal / be read by computer	B1 B1

9.

Question	Answer	Marks
3(a)	all the water is heated or the water is mixed up or water heated uniformly or distributes heat (better)	B1
	heated water rises or cold water sinks or convection transfers thermal energy (upwards)	B1
3(b)(i)	molecules move / vibrate <u>faster</u> / <u>more</u> kinetic energy	B1
	molecules push each other apart or molecules move apart or space between molecules increases or vibrate with greater amplitude	B1
3(b)(ii)	rises and liquids expand more (than solids)	B1

10.

Question	Answer	Marks
4(a)(i)	radiation or infrared (radiation / waves) or light	B1
4(a)(ii)	it / a black surface is a good absorber / poor reflector of radiation	B1
	<u>more</u> energy / power output or <u>more</u> electricity produced	B1
4(b)(i)	$(P =) VIt$ or $24 \times 15 \times 3600$ or $24 \times 15 \times 60$ or 22 000 (J)	C1
	1.3×10^6 J	A1
4(b)(ii)	$(\Delta Q =) mc\Delta T$ or 29 (°C) or 45 – 16 (°C)	C1
	$51 \times 4200 \times 29$ or $51 \times 4200 \times (45 - 16)$	C1
	6.2×10^6 J	A1

11.

Question	Answer	Marks
3(a)	conduction in metal or convection (in liquid) mentioned by name	B1
	conduction explained as heat / energy passing from molecule to molecule or movement / diffusion / collision of (free) electrons	B1
	convection explained by rising of hot liquid or correct density changes	B1
3(b)	air is a bad conductor or less area in contact / all of cup does not touch plate	B1
3(c)	white and shiny	B1
	less radiation emitted / less emission	B1

12.

(a) poor absorber / good reflector of (infra-red) radiation B1
(not with poor emitter)
less thermal energy absorbed B1

(b) (i) (pressure / it) decreases B1
molecules slow down B1
less frequent / less violent (molecular) collisions **with wall** B1

(ii) (pressure difference causes) a downward force on lid
or pressure outside > pressure inside B1

13.

(a) wood is a poor / not a conductor **or** (good) insulator (of heat) B1

(b) (i) vibrating atoms / ions / particles / molecules **or** electrons gain energy B1
atoms / ions / particles / molecules hit free electrons **or** electrons travel (a long distance through the copper / saucepan) B1
electrons hit / transfer energy to (distant) atoms / ions / molecules / particles B1

(ii) hot / heated water expands / is less dense B1
hot / heated water / less dense water rises B1
(sets up) circulation / convection (current) **or** cold water sinks B1

- 14.
- (a) (i) **heated / hot water** expands **or** density of **heated / hot water** decreases B1
 (heated / hot water) rises B1
 convection (current)/circulation set up **or** (heated / hot water) rises **and** cold B1
 water sinks
- (ii) convection transfers heat upwards **or** less dense / heated / hot water (already) B1
 at top
- (b) (i) $(Q =) VIt$ **or** $230 \times 9.6 \times 3.5$ **or** $230 \times 9.6 \times 3.5 \times 60$ **or** 7728 C1
 $4.6(368) \times 10^5 \text{ J}$ A1
- (ii) $(\Delta T =) Q / mc$ **or** $4.6(3680) \times 10^5 / 1.6 \times 4200$ C1
 69 (°C) C1
 91 °C A1
- (iii) evaporation **or** thermal energy / heat in **plastic**
casing / element / surroundings (i.e. air or environment) B1
- (c) (i) poor conductor (heat or electricity) **or** less heat lost / cooler to touch **or** less B1
 risk of shock
- (ii) poor emitter **and** less heat lost / of radiation / IR (**not** poor absorber) B1
- (d) (i) **temperature** where liquid and vapour / gas coexist **or** where liquid (**not** B1
 substance) boils
 (at atmospheric pressure)(**allow** becomes vapour / gas)
- (ii) (work done) against / overcoming forces between molecules **or** molecules B1
 gain P.E. (**ignore** K.E. increases) B1
 changes to P.E. / molecules separate

15.

- (a) (i) (conduction occurs) through or in metal/pan **or** from water to metal/pan
or molecules vibrate **or** molecules collide
or (free) electrons (in metal) move [B1]
- vibration/energy/heat passed **from molecule to molecule** clear
or energy passed on by electrons colliding (with atoms/molecules or electrons) [B1]
- (ii) hot air or air over water rises or hot water rises [B1]
hot air or hot water expands **or** hot air or water less dense [B1]
- (b) (i) black objects radiate heat more (than white) [B1]
- (ii) (both) graphs higher (after start)
or temperature falls less (in same time)/slower
or takes longer to cool [B1]
- less **evaporation** occurs **or** less **convection** [B1]
- (c) (i) heat/energy to change the temperature by 1°C/unit temp [C1]
- heat/energy to change the temperature of 1 kg/unit mass by 1°C/unit temp [A1]
- (ii) long time to warm/boil water/cook
or scalds/burns when touched
or more energy needed (to warm water) [B1]
- (iii) 1. 34(°C) **or** 94–60 seen [C1]
(m=) $Q/c\Delta T$ algebraic or numerical with any clear Q or ΔT [C1]
0.5(042) kg [A1]
2. $0.50 \times 4200 \times 54$ [C1]
110 000 **or** 114(353) J [A1]

16.

- (a) (energy transmitted) by electromagnetic/infra-red (wave)/can travel
through a vacuum B1
infra-red **or** visible $< \lambda <$ microwaves **or** λ just longer than visible
(i.e. infra-red scores 2/2) B1
- (b) (i) air is a poor conductor B1
- (ii) convection occurs (primarily) upwards/hot air rises (**not** heat rises) B1

- 17.
- (a) 16×7.5 or 120 or 96–17 or 79 C1
 $(Q =)mc\Delta T$ or $120 \times 2300 \times 79$ C1
 $2.2(2.1804) \times 10^7 \text{ J}$ A1
- (b) (i) $2.2 \times 10^7/7$ or $2.2 \times 10^7/(7 \times 60)$ or $2.2 \times 10^7/(7 \times 3600)$ C1
 $3.1 \times 10^6 \text{ J/h}$ or $5.2 \times 10^4 \text{ J/min}$ or 870 J/s or W A1
- (ii) (heater/bricks) hot(ter) (**not** room cooler) B1
great(er) temperature difference (between heater and room) B1
- (c) air (next to heater) gets hot or conduction through metal/casing B1
expands or radiation or IR (radiation) B1
less dense B1
rises B1
circulation or convection current or arrows on Fig. 10.2 B1
- (d) double glazing/cavity walls/ceiling
tiles/carpet/curtains/loft insulation etc. or shiny foil B1
traps air radiation reflected M1
air is poor conductor/convection IR radiation/
prevented back into room A1
- 18.
- (a) space is a vacuum/empty B1
these methods need matter/medium/molecules
or do not occur in vacuum B1
- (b) any **three** of:
- day:** white is a poor absorber/good reflector
- day:** less heat absorbed/less heating (of house)
- night:** white is a poor emitter/radiator
- night:** less heat emitted/heat loss (from house)
- anywhere:** of IR/radiation/radiant heat B3

19.	(a) large(r) temperature difference (between bedroom and outside) OR outside is hot(ter than main room)	B1
(b)	(i) 3 300 000 J(/hour) (ii) $(E =) P \times t$ in any form; $300 \times 60 \times 60$ $1.08 \times 10^6 \text{ J}$; $1.1 \times 10^6 \text{ J}$ OR 0.3 kWh	B1 C1 A1
(c)	cold air sinks (cold air has a) high(er) density or contracts hot air rises OR hot air has a low(er) density OR (hot) air comes in to replace cold air	B1 B1 B1
20.	(a) radiation or infra-red or electromagnetic waves travels through space/vacuum or does not require medium/molecules/particles or medium required for conduction and/or convection or for other methods (b) conduction occurs or atoms/particles/molecules vibrate or electrons given energy heat/energy/vibration passed on from one particle to another or electrons move to other parts/diffuse/hit atoms (c) $(Q =) mcT$ algebraic or numerical in any form (e.g. $1.2 \times 10^6 = m \times 400 \times 20$) 150 kg	B1 B1 B1 B1 C1 A1
21.	(a) (i) conduction (ii) molecules hit each other or molecules pass vibration on or free electrons move (through metal) and hit molecules (b) (i) downwards at or near X (ii) hot water less dense or cold water more dense hot water rises (not heat rises) or cold water falls convection current mentioned or water flows to replace hot water that rises or rising and falling described or water cools at surface	B1 B1 B1 B1 B1

- 22.
- (a) (i) **molecules/atoms/particles** escape/leave **or** liquid **molecules** change to gas/vapour B1
fastest/high energy molecules evaporate/energy needed to break bonds/latent heat B1
- (ii) hot air less dense **or** cold air more dense **or** air expands **or** body heat **conducted** into air B1
- (b) trapped air
air is a bad conductor/good insulator
convection **current** reduced **or** (air) **flow** reduced
(shiny) heat/IR/radiation reflected **or shiny** less radiation/heat emitted
evaporation reduced/air more humid, etc. ANY 3 lines 1 each B3
- 23.
- (a) good absorber (**not** good absorber and emitter) (not attracts) B1
radiation **or** infra red (not heat) B1
- (b) hot water rises (not heat rises) B1
by convection (currents) **or** density explanation B1
- (c) (i) reduce/avoid/prevent loss of heat B1
- (ii) cover/wrap in lagging/any sensible material (**not** wood/insulation, **acc.** plastic tank) B1
- 24.
- (a) time **or** observe when wax melts/falls **or** states first to melt/fall B1
first to do so **or** less wax left (after given time) (transfers heat best) B1
- (b) black **or** black cools quickly **M1**
better emitter (of heat) A1 **OR** better radiator/black radiates white doesn't A1
radiation/infra-red A1 of heat/infra-red A1
Accept in terms of white teapot (NOT better emitter and absorber/conductor)
- 25.
- (a) (i) radiation B1
(ii) no molecules or medium (to vibrate, conduct, convect) / vacuum B1
- (b) **hot air** rises B1
(hot) air expands / density decreases B1
- (c) fiberglass or air is a bad conductor/ insulator / lags / reduces heat flow
fiberglass traps air or prevents convection B1
(ignore radiation statements) B1